

Case Study: Breaking the 0.1% Barrier—FEAAM's Neural Network Innovation in Electric Drive Efficiency

Client: Electric Vehicle (EV) Manufacturers & Aerospace Propulsion Firms

Prepared by: FEAAM GmbH

Subject: Replacing Static Control Logic with Neural Networks for Breakthrough Electric Drive Efficiency

Executive Summary

In the high-stakes race for global electrification, manufacturers have hit a "mathematical plateau" where incremental hardware changes offer diminishing returns. FEAAM GmbH has broken this barrier by replacing traditional, static control logic with a Neural Network (NN)-based controller. By capturing complex motor behaviors that standard software ignores, this innovation achieved a 0.65% total energy reduction over a full WLTP driving cycle. In an industry where a 0.1% gain is considered a major milestone, this represents a generational leap in drive system intelligence.

The Challenge: The "Hidden Tax" of Static Logic

Conventional motor control relies on Lookup Tables (LUTs) - fixed data maps that tell the motor how to react. This creates three strategic "taxes" for performance:

- **The Memory Tax:** As you add more variables like temperature or battery voltage, LUTs become massive, straining the motor's computer memory.
- **The Accuracy Tax:** Traditional formulas often "guess" or oversimplify magnetic nonlinearities and iron losses, especially at high speeds.
- **The Efficiency Tax:** These limitations force engineers to use "nominal" settings, leaving significant energy on the table during real-world driving.

The FEAAM Innovation: Neural Network-Driven Control

FEAAM eliminates these trade-offs by deploying a compact, 12-layer Neural Network directly into the control loop.

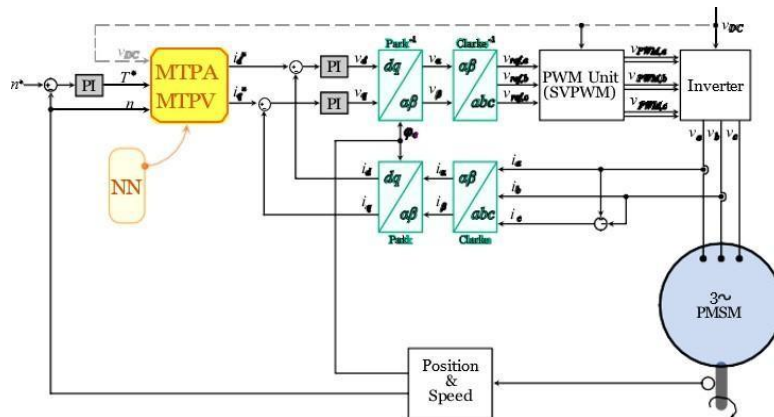


Fig. 1: Field-Oriented Control structure with NN replacing conventional MTPA/MTPV block

How it Works:

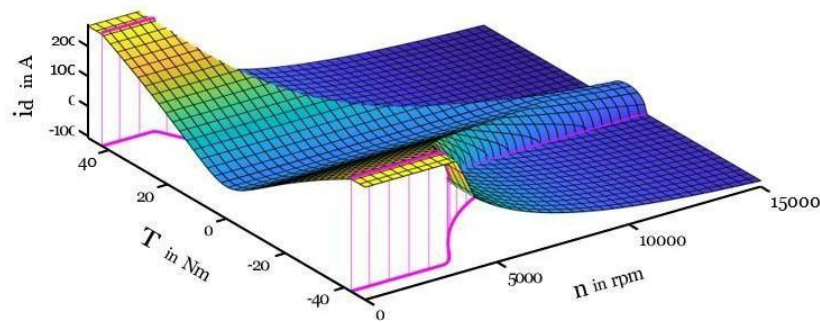


Fig. 2: Optimal current surface i_d^* computed via Constrained Minimization Algorithm (CMA)

- Ground Truth Modeling: FEAAM builds machine models from actual test bench measurements to capture real-world iron losses.
- Mathematical Filtering: A "Constrained Minimization Algorithm" solves millions of optimization problems to find the absolute points of minimum energy loss.
- Intelligent Reconstruction: The NN acts as a high-speed "brain," remembering these millions of points in a compact format (only 1,000 weights) that fits on standard, low-cost hardware.
- Real-Time Speed: With an inference time of just 0.5 ms, the system reacts with surgical precision in real-time.

Proof of Performance: The 0.65% Breakthrough

FEAAM validated this approach against the WLTP cycle, the global standard for simulating real-world driving.

- **0.65% Total Energy Reduction:** The NN-based approach outperformed the industry-standard LUTs across the entire test cycle.
- **4% Peak Local Gains:** In specific high-speed regions, the NN identified efficiency savings that traditional logic simply could not "see".
- **Beyond Physical Limits:** While "Machine 1" was limited by heat, "Machine 2" (representative of aircraft propellers) proved the NN's superiority in high-loss, high-speed conditions.

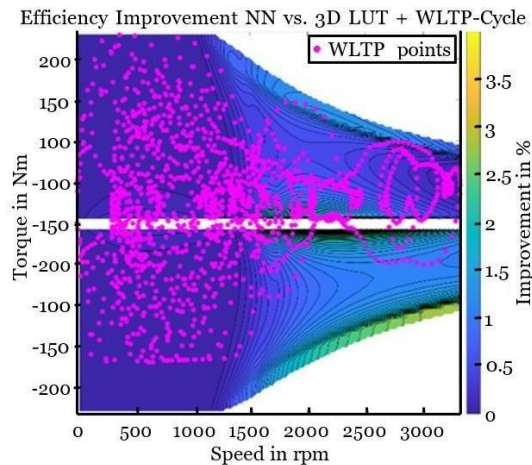


Fig. 3. WLTP operating points mapped onto the efficiency improvement surface for Machine 2

The Bottom Line for Business

This software-defined evolution provides a massive ROI for high-volume manufacturers:

- **Direct Cost Avoidance:** Higher efficiency allows for reduced battery capacity while maintaining the same range, lowering the vehicle's total cost.
- **Fleet-Wide Impact:** A 0.65% gain saves several megawatt-hours over a fleet's lifetime, drastically reducing total energy costs and carbon footprints.
- **Infinite Scalability:** Unlike LUTs, the NN handles new variables (like aging or flux variations) without "crashing" the system memory.

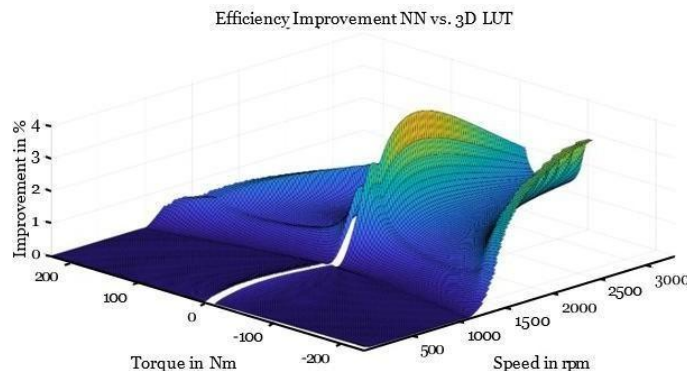


Fig. 4. Operating points from WLTP cycle mapped on NN-generated efficiency surface (Machine 2)

FEAAM can assist your team by integrating this intelligent control architecture into your existing development cycle, capturing energy gains in systems previously considered "fully optimized".